Project List 2

33-467: Astrophysics of the Stars and Galaxy

Fall 2023

1 Orbital Evolution of Binaries in Roche-lobe Overflow

This project has both analytic and computational components.

2 Finding black holes with Gaia

Gaia BH1 paper

3 The Wild World of Binary-Star Evolution

These projects will use the population synthesis code COSMIC which simulates single and binary-star populations at phases across the H-R Diagram through to the formation of stellar remnants. There are *many* different uncertainties associated with binary evolution due to different interactions between binary components that can occur based on the initial properties of the system and assumptions made for how each interaction should be treated. This means that there are *a lot* of opportunities for exploration. You should explore the COSMIC documentation and COSMIC code paper for a summary of the different assumptions made in the code as well as the Binary Evolution in a Nutshell notes to find an aspect of binary star interactions that you find most interesting. Maybe it's studying how different assumptions for wind mass loss affects binary populations? Maybe it's investigating how compact object populations depend on the strength of supernova natal kicks? Maybe it's something else!

Whatever topic you choose should not be exactly the same as any other projects, so you may need to coordinate with your colleagues. Once you have a topic to study, you should determine which assumptions are set by flags in COSMIC that affect that topic as well as determine which binary populations will show the largest changes by using different assumptions. You should then summarize how you expect each assumption you make to alter the binary population from its initial configuration at ZAMS. Finally, you should evolve large populations of binaries with COSMIC using the flags you've chosen and make figures which illustrate how each assumption impacts the population.

4 Investigating Dark Matter with Stellar Streams

5 The Fastest Stars in the Galaxy

6 Connecting Dust Morphology to Star Formation

7 Kinematics of Compact Objects in the Galaxy

8 Chemical Evolution with the Cannon

9 Exoplanet Orbital Dynamics with Rebound